

Hyperbolic geometry, discrete groups, and manifolds

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The hyperbolic plane

The goal of the first chapter is to introduce the main protagonist of this course, the hyperbolic plane. We will see some historical motivation, and define the hyperbolic plane using several different models. The multitude of available models will enrich our study of hyperbolic geometry, and we will learn to jump from one to the other depending on the situation. The formal definitions of the models for the hyperbolic plane rely on Riemannian geometry; the necessary notions will also be introduced in this chapter.

Recommended reading:

- [Cannon et al., 1997, §1–8]
- [Thurston, 1997, §2.1–3]
- [Beardon, 1983, §7.1–2]

LECTURE 1 — THE POINCARÉ HALF-PLANE

1.1 Non-Euclidean geometry

Our journey into hyperbolic geometry begins with Euclidean geometry, or rather its negation. Euclidean geometry is the geometry of the Euclidean plane (or more generally, of Euclidean n -space). It was axiomatised by Euclid.

Axioms I.1 (Euclid). (i) *Every pair of points can be joined by a unique straight line segment.*
 (ii) *Every straight line segment can be extended indefinitely in both directions.*
 (iii) *There is a unique circle of a given centre and radius.*
 (iv) *All right angles are congruent to one another.*
 (v) *Given a line ℓ and a point $p \notin \ell$, there is a unique line ℓ' containing p and parallel to ℓ .*

The fifth axiom is sometimes also called the *parallel postulate*¹. As the parallel postulates sounds more complicated than the other four axioms, mathematicians have

¹In fact, Euclid's fifth axiom was a little different, but equivalent to the parallel postulate given the other four. See [Cannon et al., 1997].

spent a lot of time and energy trying to prove that it can be deduced from them, in order to simplify the axioms of geometry (at this stage, there is no need to specify “Euclidean”, since this is the only geometry that was known).

In the nineteenth century, several mathematicians (including Gauß, Lobachevskiĭ, Bolyai) had the fecund idea of trying to create a consistent theory of geometry where the parallel postulate does not hold — this gave birth to *hyperbolic geometry*, which was the first non-Euclidean geometry to be discovered and explored. Later, Riemann introduced what is now called *Riemannian geometry*, which provides a very general setup to define and study a very broad array of geometries. The Riemannian setup now gives us a very convenient language to define hyperbolic geometry, and we will take the liberty to follow this anachronism.

Ideas from Riemannian geometry will already appear in this lecture without general definitions; they will be formalised in the second lecture.

1.2 Definition of the Poincaré half-plane

The underlying set of our first model is the complex upper half-plane:

$$\mathbb{H}^2 := \{z \in \mathbb{C} : \Im(z) > 0\}.$$

For each point $z \in \mathbb{H}^2$, we define a *local hyperbolic norm* $\|\cdot\|_{\text{hyp}}^z$ via

$$\|v\|_{\text{hyp}}^z := \frac{1}{\Im(z)} \|v\|_{\text{eucl}},$$

for all $v \in \mathbb{R}^2$, where $\|\cdot\|_{\text{eucl}}$ is the usual Euclidean norm on $\mathbb{R}^2$. Defining local norms at each point of our space might sound a little strange for now, but this will be clarified when we introduce Riemannian geometry. For now, the only thing we need is that this allows us to define hyperbolic lengths and distances.

Definition I.2. The *hyperbolic length* of a (piecewise) C^1 path $\gamma : [a, b] \rightarrow \mathbb{H}^2$ is

$$|\gamma|_{\text{hyp}} := \int_a^b \|\gamma'(t)\|_{\text{hyp}}^{\gamma(t)} dt.$$

The *hyperbolic distance* between two points $z, w \in \mathbb{H}^2$ is

$$d_{\text{hyp}}(z, w) := \inf \left\{ |\gamma|_{\text{hyp}} : \gamma \in C_{\text{pw}}^1([a, b], \mathbb{H}^2), \gamma(a) = z, \gamma(b) = w \right\},$$

where $C_{\text{pw}}^1([a, b], \mathbb{H}^2)$ is the set of piecewise C^1 paths $[a, b] \rightarrow \mathbb{H}^2$.

Exercise 1.A

Let γ be a path of the form

$$\gamma : t \mapsto x_0 + iy(t)$$

for some $x_0 \in \mathbb{R}$ and some strictly increasing piecewise C^1 function $y : [a, b] \rightarrow (0, +\infty)$. Compute $|\gamma|_{\text{hyp}}$.

Exercise 1.B

Let $\gamma : [a, b] \rightarrow \mathbb{H}^2$ be a piecewise C^1 path, and let $\phi : [a', b'] \rightarrow [a, b]$ be a strictly increasing C^1 function. Prove that $|\gamma \circ \phi|_{\text{hyp}} = |\gamma|_{\text{hyp}}$.

Proposition I.3. d_{hyp} defines a metric on $\mathbb{H}^2$.

Proof. Symmetry and the triangle inequality are clear (reverse and concatenate paths). For separation, fix $z \neq w \in \mathbb{H}^2$, and let $\gamma : [a, b] \rightarrow \mathbb{H}^2$ be a piecewise C^1 path from z to w . Decompose γ as $\gamma = x + iy$, with $x, y : [a, b] \rightarrow \mathbb{R}$, and $y(t) > 0$ for all $t \in [a, b]$.

First assume that $y(t) \leq 2y(a)$ for all $t \in [a, b]$. Then we have

$$|\gamma|_{\text{hyp}} = \int_a^b \frac{\sqrt{x'(t)^2 + y'(t)^2}}{y(t)} dt \geq \int_a^b \frac{\sqrt{x'(t)^2 + y'(t)^2}}{2y(a)} dt = \frac{1}{2\Im(z)} |\gamma|_{\text{eucl}},$$

where $|\gamma|_{\text{eucl}}$ denotes the *Euclidean length* of γ (given by $|\gamma|_{\text{eucl}} := \int_a^b \|\gamma'(t)\|_{\text{eucl}} dt$). But the Euclidean length of a piecewise C^1 path is at least equal to the Euclidean distance between its endpoints, so that

$$|\gamma|_{\text{hyp}} \geq \frac{d_{\text{eucl}}(z, w)}{2\Im(z)}.$$

Otherwise, let $t_0 := \min \{t \in [a, b] : y(t) \geq 2y(a)\}$, and let $\gamma' := \gamma|_{[a, t_0]}$. The previous computation gives

$$|\gamma|_{\text{hyp}} \geq |\gamma'|_{\text{hyp}} \geq \frac{1}{2\Im(z)} |\gamma'|_{\text{eucl}}.$$

But the Euclidean length of γ' is at least equal to the Euclidean distance between z and the line $\{u \in \mathbb{C} : \Im(u) = 2\Im(z)\}$. The latter distance is equal to $\Im(z)$, yielding $|\gamma|_{\text{hyp}} \geq \frac{1}{2}$ in this case.

In general, we have proved that

$$d_{\text{hyp}}(z, w) \geq \frac{1}{2} \min \left\{ 1, \frac{d_{\text{eucl}}(z, w)}{\Im(z)} \right\} > 0. \quad \square$$

The metric space $(\mathbb{H}^2, d_{\text{hyp}})$ is called the *Poincaré half-plane*.

Exercise 1.C

What is the topology on $\mathbb{H}^2$ induced by d_{hyp} ?

1.3 First examples of shortest paths and isometries

The following will give us shortest paths in $\mathbb{H}^2$ in a very specific case.

Lemma I.4. *Consider the orthogonal projection $p : \mathbb{H}^2 \rightarrow \mathbb{H}^2$ given by $z \mapsto z - \Re(z)$. Given a piecewise C^1 path $\gamma : [a, b] \rightarrow \mathbb{H}^2$, we have*

$$|p \circ \gamma|_{\text{hyp}} \leq |\gamma|_{\text{hyp}},$$

with equality if and only if $\Re(\gamma)$ is constant.

Proof. Write $\gamma = x + iy$, where $x, y : [a, b] \rightarrow \mathbb{R}$, with $y(t) > 0$ for all $t \in [a, b]$. We have

$$|p \circ \gamma|_{\text{hyp}} = \int_a^b \frac{|y'(t)|}{y(t)} dt \leq \int_a^b \frac{\sqrt{x'(t)^2 + y'(t)^2}}{y(t)} dt = |\gamma|_{\text{hyp}},$$

with equality if and only if $x'(t) = 0$ for all $t \in [a, b]$. $\square$

Proposition I.5. *Let $z, w \in \mathbb{H}^2$.*

- (i) *We have $d_{\text{hyp}}(z, w) \geq |\log \Im(z) - \log \Im(w)|$.*
- (ii) *If $\Re(z) = \Re(w)$, then the vertical segment is the unique (up to reparametrisation) shortest path between z and w .*

Proof. By Lemma I.4, it suffices to prove (i) when $\Re(z) = \Re(w)$. Moreover, it suffices to consider piecewise C^1 paths γ from z to w for which $\Re(\gamma)$ is constant. The computation of Exercise 1.A shows that the length of such a path is at least $|\log \Im(z) - \log \Im(w)|$, with equality if and only if $\Im(\gamma)$ is strictly increasing or decreasing. Hence, $d_{\text{hyp}}(z, w) \geq |\log \Im(z) - \log \Im(w)|$, and this is attained exactly by reparametrisations of the vertical segment from z to w . $\square$

Isometries will be key in many geometric arguments throughout the course. The following observation allows us to easily verify that certain maps are isometries; the hypothesis will turn out to be the definition of a Riemannian isometry, which we will see in the next lecture.

Observation I.6. Let $f : \mathbb{H}^2 \rightarrow \mathbb{H}^2$ be a C^1 diffeomorphism such that, for every $z \in \mathbb{H}^2$ and for every $v \in \mathbb{R}^2$,

$$\|df_z(v)\|_{\text{hyp}}^{f(z)} = \|v\|_{\text{hyp}}^z.$$

Then f is an isometry: for all $z, w \in \mathbb{H}^2$,

$$d_{\text{hyp}}(f(z), f(w)) = d_{\text{hyp}}(z, w).$$

Proof. It is easy to verify that $|f \circ \gamma|_{\text{hyp}} = |\gamma|_{\text{hyp}}$ for every piecewise C^1 path γ ; it follows that f preserves d_{hyp} . $\square$

Exercise 1.D

Prove that the following maps are isometries of $\mathbb{H}^2$:

- (i) Horizontal translations,
- (ii) Reflections along vertical lines,
- (iii) Homotheties.

Deduce that the action of the isometry group $\text{Isom}(\mathbb{H}^2) \curvearrowright \mathbb{H}^2$ is transitive.

In fact, we will see later that the action $\text{Isom}(\mathbb{H}^2) \curvearrowright \mathbb{H}^2$ is transitive in a stronger sense: $\text{Isom}(\mathbb{H}^2)$ acts transitively on pairs of points at a given distance. Together with Proposition I.5, this will allow us to deduce that any two points in $\mathbb{H}^2$ are connected by a unique shortest path. This is a typical example of the use of isometries to deduce geometric properties of the hyperbolic plane by reducing to an easier case; we will see many arguments of this kind.

Exercise 1.E

The hyperbolic line is the set $\mathbb{H}^1 := (0, +\infty)$, with local hyperbolic norm given by $\|v\|_{\text{hyp}}^y = \frac{|v|}{y}$ for $y \in \mathbb{H}^1$ and $v \in \mathbb{R}$. The hyperbolic length of a (piecewise) C^1 path and the hyperbolic distance are defined analogously to the 2-dimensional case.

- (i) Compute $d_{\text{hyp}}(y_1, y_2)$ for $y_1, y_2 \in \mathbb{R}$, and deduce that $\mathbb{H}^1$ admits an isometric embedding into $\mathbb{H}^2$.
- (ii) Show that $\mathbb{H}^1$ is isometric to $\mathbb{R}$ with the usual norm.
- (★) Is $\mathbb{H}^2$ isometric to $\mathbb{R}^2$ with the Euclidean norm?

LECTURE 2 — RIEMANNIAN GEOMETRY, THE POINCARÉ DISC

This lecture introduces the basics of Riemannian geometry that will be useful for defining our different models of the hyperbolic plane; many propositions and theorems here will be left unproved. We work in the smooth category throughout.

2.1 Definition of a Riemannian manifold

Let M be a (smooth) d -manifold ($d \geq 1$). Recall that the *tangent bundle* of M is a $2d$ -manifold decomposing as

$$TM = \coprod_{p \in M} T_p M,$$

where, for each $p \in M$, the subspace $T_p M$ is the tangent space at p , and carries the extra structure of a real vector space of dimension d . The tangent bundle is equipped with a smooth projection $\pi : TM \rightarrow M$.

The *2-covariant tensor bundle* of M is a manifold decomposing as

$$(TM^*)^{\otimes 2} = \coprod_{p \in M} T_p M^* \otimes T_p M^*,$$

where $T_p M^* := \text{Hom}_{\mathbb{R}}(T_p M, \mathbb{R})$ is the (linear) dual of $T_p M$. For each $p \in M$, we will identify elements of $T_p M^* \otimes T_p M^*$ with bilinear forms $T_p M \times T_p M \rightarrow \mathbb{R}$. Again, $(TM^*)^{\otimes 2}$ is equipped with a smooth projection $\pi : (TM^*)^{\otimes 2} \rightarrow M$.

Definition I.7. A *Riemannian manifold* is the data (M, g) of a (smooth) manifold M together with a map $g : M \rightarrow (TM^*)^{\otimes 2}$ which

- is a section of $\pi : (TM^*)^{\otimes 2} \rightarrow M$ — i.e. $\pi \circ g = \text{id}_M$ — and
- induces a scalar product at each point — i.e. for each $p \in M$, the bilinear form $g(p) : T_p M \times T_p M \rightarrow \mathbb{R}$ is a scalar product.

The map g is called a *Riemannian metric* (or Riemannian structure) on M .

We have not defined the smooth structures on the tangent and tensor bundles, and we will not really need the above technical language. The important idea to keep in mind is that a Riemannian metric on a manifold M is the data of a scalar product $g(p)$ on each tangent space $T_p M$, such that $g(p)$ varies smoothly with p . We will denote the scalar product $g(p)$ on $T_p M$ by $\langle \cdot, \cdot \rangle_g^p$, and the associated norm by $\| \cdot \|_g^p$: for each tangent vector $v \in T_p M$,

$$\|v\|_g^p := \sqrt{\langle v, v \rangle_g^p}.$$

Example I.8. Let M be an embedded d -submanifold of $\mathbb{R}^N$. Equip $\mathbb{R}^N$ with its standard Euclidean structure. For each $x \in M$, the tangent space $T_x M$ can be seen as a linear subspace of $\mathbb{R}^N$; let $g(x)$ be the restriction of the scalar product on $\mathbb{R}^N$ to $T_x M$. Then g defines a Riemannian metric on M .

More generally, an embedded submanifold M of a Riemannian manifold M_0 inherits a Riemannian structure from M_0 , called the *induced Riemannian structure*.

Let $(x_1, \dots, x_d)$ be local coordinates for the d -manifold M at a point p . We normally denote by $(dx_1, \dots, dx_d)$ the basis of $T_p M^*$ which is dual to the basis $(e_1, \dots, e_d)$ of $T_p M$ induced by the local coordinates. The unique scalar product $\langle \cdot, \cdot \rangle$ on $T_p M$ making $(e_1, \dots, e_d)$ an orthonormal basis is then denoted by

$$dx_1^2 + \dots + dx_d^2.$$

These notations will be convenient for defining Riemannian metrics, and we will use them repeatedly in this chapter.

Example I.9. The Poincaré half-plane is the Riemannian surface (2-manifold) whose underlying manifold is $\mathbb{H}^2 := \{(x, y) \in \mathbb{R}^2 : y > 0\}$ with its usual smooth structure, with the Riemannian metric

$$\frac{dx^2 + dy^2}{y^2}.$$

Compare with §1.2.

Exercise 2.A

Consider the sphere

$$\mathbb{S}^2 := \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 + z^2 = 1\} \subseteq \mathbb{R}^3.$$

Write the Riemannian metric on $\mathbb{S}^2$ induced by the standard Riemannian structure of $\mathbb{R}^3$ in the local coordinates given by stereographic projection.

2.2 Lengths of paths, distances, and isometries

For the rest of this lecture, we fix a (connected) Riemannian manifold (M, g) .

Definition I.10. Let $\gamma : [a, b] \rightarrow M$ be a (piecewise) C^1 path. The *length* of γ is

$$|\gamma|_g := \int_a^b \|\gamma'(t)\|_g^{\gamma(t)} dt.$$

The *distance* between two points $p, q \in M$ is

$$d_g(p, q) := \inf \left\{ |\gamma|_g : \gamma \in C_{\text{pw}}^1([a, b], M), \gamma(a) = p, \gamma(b) = q \right\}.$$

Remark I.11. The function $d_g : M \times M \rightarrow [0, +\infty)$ defines a metric on M (as for Proposition I.3, separation is not obvious but will follow from the results appearing later in this lecture). The topology on M induced by d_g coincides with the topology on M as a manifold; the proof is the same as that of Exercise 1.C.

Definition I.12. Let (M, g) and (N, h) be two Riemannian manifolds. A *Riemannian isometric embedding* $f : (M, g) \hookrightarrow (N, h)$ is a smooth embedding $f : M \hookrightarrow N$ which preserves the Riemannian metrics in the sense that, for all $p \in M$ and $v, w \in T_p M$, we have

$$\langle df_p(v), df_p(w) \rangle_h^{f(p)} = \langle v, w \rangle_g^p.$$

A *Riemannian isometry* is a Riemannian isometric embedding which is in addition a diffeomorphism.

Proposition I.13. If $f : (M, g) \rightarrow (N, h)$ is a Riemannian isometry, then f is also an isometry of metric spaces $(M, d_g) \rightarrow (N, d_h)$.

Proof. See Observation I.6. □

Example I.14. The d -torus $\mathbb{T}^d := \mathbb{R}^d / \mathbb{Z}^d$ has a Riemannian structure defined in the local coordinates $(x_1, \dots, x_d)$ coming from $\mathbb{R}^d$ by $dx_1^2 + \dots + dx_d^2$. This is called the *flat structure* on $\mathbb{T}^d$.

For $d = 2$, note that there is no (smooth) Riemannian isometric embedding $\mathbb{T}^2 \hookrightarrow \mathbb{R}^3$; in particular, the above-defined Riemannian structure on $\mathbb{T}^2$ is different from any metric obtained by smoothly embedding $\mathbb{T}^2$ (as a manifold) into $\mathbb{R}^3$.

Exercise 2.B

Prove that the natural projection $\mathbb{R}^d \rightarrow \mathbb{T}^d$ is a Riemannian isometric embedding for the flat structure on $\mathbb{T}^d$. Is it an isometric embedding of metric spaces?

The following theorem is far beyond the scope of this course, but we mention the result.

Theorem I.15 (Nash Embedding). If (M, g) is a d -dimensional Riemannian manifold, then there exist $D \leq \frac{1}{2}d(d+1)(3d+11)$, as well as a Riemannian isometric embedding $M \hookrightarrow \mathbb{R}^D$.

2.3 Geodesics

Geodesics are important paths in Riemannian manifolds. In a course on Riemannian geometry, they would usually be defined as local minimisers of the *energy functional*

$$\mathcal{E} : \gamma \mapsto \int_a^b \left(\|\gamma'(t)\|_g^{\gamma(t)} \right)^2 dt.$$

Such local minimisers can be characterised by a differential equation depending on the Riemannian metric; this is called the *equation of geodesics*. For our purpose, the following equivalent definition will however suffice.

Definition I.16. A *geodesic* on (M, g) is a C^1 path $\gamma : I \rightarrow M$ (where I is an interval) for which there is a constant $K > 0$ satisfying:

- (i) (Constant speed) $\|\gamma'(t)\|_g^{\gamma(t)} = K$ for all $t \in I$,
- (ii) (Local length minimisation) For each $t_1 \in I$, there is an open subinterval J of I containing t_1 such that, for all $s, t \in J$,

$$d_g(\gamma(s), \gamma(t)) = \left| \gamma|_{[s,t]} \right|_g = K|s - t|.$$

Exercise 2.C

Let $f : (M, g) \rightarrow (N, h)$ be a Riemannian isometric embedding. Prove that, if $\gamma : I \rightarrow M$ is a geodesic in (M, g) , then $f \circ \gamma$ is a geodesic in (N, h) . Deduce that f is a local isometric embedding of metric spaces.

Proving the following theorem would require going via the equation of geodesics; we will merely remember the result.

Theorem I.17. Let (M, g) be a Riemannian manifold. For every $p \in M$ and $v \in T_p M$, there is an open interval I containing 0 and a geodesic $\gamma : I \rightarrow M$ such that

$$\gamma(0) = p \quad \text{and} \quad \gamma'(0) = v.$$

Such a geodesic with the interval I maximal for inclusion is unique.

It now follows from Proposition I.5(ii) that, in $\mathbb{H}^2$, the vertical ray $i\mathbb{R}_{>0}$ is a geodesic (with an appropriate parametrisation).

2.4 The exponential map

We now turn to the last notion we will need from Riemannian geometry: the exponential map.

Definition I.18. Given $p \in M$, let U_p be the set of tangent vectors $v \in T_p M$ for which the maximal geodesic $\gamma_{(p,v)} : I_{(p,v)} \rightarrow M$ satisfies $1 \in I_{(p,v)}$. The *exponential map* at p is the smooth map

$$\begin{aligned} \exp_p : U_p &\longrightarrow M \\ v &\longmapsto \gamma_{(p,v)}(1). \end{aligned}$$

Proposition/Definition I.19. Let (M, g) be a Riemannian manifold, and let $p \in M$. There is $\varepsilon > 0$ such the map

$$\exp_p : B(0, \varepsilon) \rightarrow M$$

is injective (and in fact, is a diffeomorphism onto its image). Here, we write $B(0, \varepsilon) := \{v \in T_p M : \|v\|_g^p < \varepsilon\}$.

The supremum of such ε is called the *injectivity radius* of M at p , and denoted by $\text{inj}_p(M)$. We also define the (global) *injectivity radius* of M by

$$\text{inj}(M) := \inf_{p \in M} \text{inj}_p(M).$$

Exercise 2.D

If $\gamma : [0, 1] \rightarrow M$ is a geodesic with $\gamma(0) = \gamma(1) = p$, prove that

$$\text{inj}_p(M) \leq \frac{1}{2} |\gamma|_g.$$

Exercise 2.E

Let (M, g) be a Riemannian manifold.

- (i) Let $\gamma : [a, b] \rightarrow M$ be a C^1 path with constant speed (i.e. there is a constant K such that $\|\gamma'(t)\|_g^{\gamma(t)} = K$ for all $t \in [a, b]$). Suppose that for every C^1 path $\eta : [a, b] \rightarrow M$ with $\gamma(a) = \eta(a)$ and $\gamma(b) = \eta(b)$, we have $|\gamma|_g \leq |\eta|_g$. Prove that γ is a geodesic.
- (ii) Prove that the converse does not hold.
- (iii) Prove the following partial converse: let $p \in M$, and consider $B := \{v \in T_p M : \|v\|_g^p < \text{inj}_p(M)\}$. If $\gamma : [0, 1] \rightarrow \exp_p(B)$ is a geodesic with $\gamma(0) = p$, and $c : [0, 1] \rightarrow M$ is a piecewise C^1 path with $c(0) = \gamma(0)$ and $c(1) = \gamma(1)$, then

$$|c|_g \geq |\gamma|_g.$$

2.5 Completeness and the Hopf–Rinow Theorem

We finish by mentioning the following important theorem.

Theorem I.20 (Hopf–Rinow). *Given (M, g) a Riemannian manifold and $p \in M$, the following assertions are equivalent:*

- (i) $\exp_p$ is defined on all of $T_p M$.
- (ii) Closed and bounded subsets of M are compact.
- (iii) M is Cauchy-complete.
- (iv) M is geodesically complete: every maximal geodesic is defined over $\mathbb{R}$.

Moreover, the above conditions imply:

- (v) For all $q \in M$, there is a geodesic $\gamma : \mathbb{R} \rightarrow M$ with $\gamma(0) = p$, $\gamma(1) = q$, and $|\gamma|_{0,1}|_g = d_g(p, q)$.

Example I.21. The Euclidean space $\mathbb{R}^d$, the d -torus $\mathbb{T}^d$, the d -sphere $\mathbb{S}^d$, are all geodesically complete.

Remark I.22. The implication (v) $\Rightarrow$ (i) in Theorem I.20 is false: for instance, consider the open disc $D^2 := \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 < 1\}$, with the Euclidean metric $dx^2 + dy^2$.

2.6 The Poincaré disc

Definition I.23. The *Poincaré disc* is the Riemannian surface $\mathbb{D}^2 := \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 < 1\}$, with Riemannian metric given by

$$\left(\frac{2}{1 - x^2 - y^2} \right)^2 (dx^2 + dy^2)$$

Proposition I.24. *Rotations centred at zero, as well as reflections along lines passing through zero, are Riemannian isometries of $\mathbb{D}^2$.*

Of all the isometries connecting the different models, the following is the most important to remember; it is an example of a Möbius transformation, and those will turn out to play a major role.

Theorem I.25. *Identify $\mathbb{D}^2$ with $\{z \in \mathbb{C} : |z|^2 < 1\}$, and consider the Cayley transform*

$$\begin{aligned} \phi : \mathbb{H}^2 &\longrightarrow \mathbb{D}^2 \\ z &\longmapsto \frac{z - i}{z + i}. \end{aligned}$$

Then ϕ is a Riemannian isometry from $\mathbb{H}^2$ to $\mathbb{D}^2$.

Proof. To check that $\phi(\mathbb{H}^2) \subseteq \mathbb{D}^2$, note that, for $z \in \mathbb{H}^2$,

$$|\phi(z)|^2 = \frac{z - i}{z + i} \cdot \frac{\bar{z} + i}{\bar{z} - i} = \frac{|z|^2 + 1 - 2\Im(z)}{|z|^2 + 1 + 2\Im(z)} < 1,$$

since $\Im(z) > 0$. Computing $d\phi$ corresponds to computing the complex derivative ϕ' , namely

$$\phi'(z) = \frac{2i}{(z+i)^2}.$$

Now fix $z \in \mathbb{H}$. Given a tangent vector v at z seen as an element of $\mathbb{C}$, we want to check that $\|v\|_{\mathbb{H}^2}^z = \|\phi'(z)v\|_{\mathbb{D}^2}^{\phi(z)}$. Hence, we want to check the equality

$$\frac{1}{\Im(z)} = \frac{2|\phi'(z)|}{1-|\phi(z)|^2},$$

which follows from the above computations.

Finally, note that ϕ has an inverse given by

$$\begin{aligned} \psi : \mathbb{D}^2 &\longrightarrow \mathbb{H}^2 \\ w &\longmapsto i \frac{1+w}{1-w}, \end{aligned}$$

showing that ϕ is surjective. Hence, ϕ is a diffeomorphism $\mathbb{H}^2 \rightarrow \mathbb{D}^2$ respecting the Riemannian metrics, so it is a Riemannian isometry. $\square$

The existence of an isometry $\mathbb{D}^2 \cong \mathbb{H}^2$ is fundamental: it allows us to translate any geometric statement known in $\mathbb{D}^2$ to $\mathbb{H}^2$, and vice versa. Sometimes, we will even move back and forth between the two models within a single proof.

Remark I.26. The image under the Cayley transform of the vertical ray $\{0\} \times (0, +\infty) \subseteq \mathbb{H}^2$ is the horizontal diameter $(-1, +1) \times \{0\} \subseteq \mathbb{D}^2$. In particular, the horizontal diameter is a shortest path of $\mathbb{D}^2$ by Proposition I.5(ii); more than that, it is the unique shortest path between any two points on it.

Proof. For $y > 0$, note that

$$\phi(iy) = \frac{y-1}{y+1}. \quad \square$$

At the end of Lecture 1, we mentioned the fact that $\text{Isom}(\mathbb{H}^2)$ acts transitively on pairs of points at a given distance, without proving it. Looking at the hyperbolic plane from the point of view of the Poincaré disc now gives us the right tools to prove this fact.

Proposition I.27. *Let $z_1, w_1, z_2, w_2 \in \mathbb{D}^2$ with $d(z_1, w_1) = d(z_2, w_2)$. Then there exists $f \in \text{Isom}(\mathbb{D}^2)$ such that $f(z_1) = z_2$ and $f(w_1) = w_2$.*

Proof. We already know that the action $\text{Isom}(\mathbb{D}^2) \curvearrowright \mathbb{D}^2$ is transitive by Exercise 1.D and the fact that $\mathbb{D}^2 \cong \mathbb{H}^2$. Hence, we can assume that $z_1 = z_2 = 0$. Now let f_1 (respectively f_2) be the unique rotation centred at 0 sending w_1 (respectively w_2) to the radius $[0, 1) \times \{0\} \subseteq \mathbb{D}^2$. Note that $f_1, f_2 \in \text{Isom}(\mathbb{D}^2)$ by Proposition I.24. Since the radius $[0, 1) \times \{0\} \subseteq \mathbb{D}^2$ is a shortest path (see Remark I.26), and since $d(0, f_1(w_1)) = d(0, w_1) = d(0, w_2) = d(0, f_2(w_2))$, we must have $f_1(w_1) = f_2(w_2)$. Now the isometry $f := f_2^{-1} \circ f_1$ satisfies $f(z_1) = z_2$ and $f(w_1) = w_2$. $\square$

Corollary I.28. *Every pair of points in $\mathbb{D}^2$ is joined by a unique shortest path.*

Proof. Let $z, w \in \mathbb{H}^2$. By Proposition I.27, there is $f \in \text{Isom}(\mathbb{D}^2)$ such that $f(z) = 0$ and $f(w) \in [0, 1) \times \{0\}$. Hence, $f(z)$ and $f(w)$ are joined by a unique shortest path (namely the straight line segment — see Remark I.26), and therefore so are z and w . $\square$

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